An AI-ready evidence base on product testing with users in agriculture

Rachel Chase[a], Brayden Youngberg[a], Marie-Angélique Laporte[c], Kauê de Sousa[b,d,*]

[a] *Bioversity International, France*
[b] *Digital Inclusion, Bioversity International, Montpellier, France*
[c] *Performance Innovation and Strategic Analysis for Impact (PISA4), Bioversity International, Montpellier, France*
[d] *Department of Agricultural Sciences, University of Inland Norway, Hamar, Norway*

Abstract

Agricultural products succeed when they meet the needs of the people who use them. Whether a new variety, seed product or food reaches farmers and consumers depends on how it performs in their fields and kitchens, what they value, and how well it copes with climate risk. Since it was first proposed in 2011, the tricot approach to testing products with farmers and consumers has produced a large body of evidence on exactly these questions (van Etten, 2011; de Sousa et al., 2024), but it is spread across journal articles, guides, reports, briefs and presentations that few people have time to search. AI assistants could close this gap, yet an assistant without curated sources may invent results, misreport numbers or stretch a finding from one crop and country into a general rule. We present an AI-ready evidence base that distils fifteen years of openly available publications on tricot and related approaches to product testing with farmers and consumers into verified markdown cards, each recording the questions a source answers, its methods and stated limitations, key numbers with their primary sources, and links to the related data and code. The repository works as a skill for any AI assistant, with explicit rules for evidence boundaries, attribution and contact consent. It is the evidence layer of a set of linked AI-ready resources, developed within the CGIAR Digital Transformation Accelerator, on published evidence, field protocols, on-farm trial data and analytical approaches for the demand-led development of agricultural products.

Keywords: demand-led product development, product testing with users, tricot, on-farm testing, citizen science, knowledge base, AI-ready documentation, FAIR principles

Suggested citation

> Chase, R.; Youngberg, B.; Laporte, M.-A.; de Sousa, K. (2026) An AI-ready evidence base on product testing with users in agriculture. Version 0.1.0. `https://github.com/AgrDataSci/tricot-knowledge-base`.

The most recent version of the knowledge base and this paper is always available at the repository link above.

*Corresponding author
Email address: `k.desousa@cgiar.org` (Kauê de Sousa)

Statement of need

Agricultural research and development is shifting from supply to demand. A new variety, seed product or food succeeds only if farmers, processors and consumers want it, and the way to know is to test products with the people who will use them, where they will use them. The tricot approach was designed for this purpose (van Etten, 2011; van Etten et al., 2019). It has been applied to more than 30 crops in more than 25 countries and extended to consumer testing and product concept testing (de Sousa et al., 2024). Combined with genomic, socio-economic and climate data, tricot rankings have improved the prediction of on-farm performance (de Sousa et al., 2021) and revealed user segments with distinct preferences (Voss et al., 2025).

This evidence is spread across dozens of documents written for readers who already know where to look. The people who need it, from breeders and product managers to seed companies, social scientists, development organizations, funders and students, increasingly ask AI assistants instead. Without curated sources, an assistant can invent a paper, quote a number from the wrong study, or turn a result from one crop and country into a general rule. Box 1 shows what changes when the assistant answers from the knowledge base.

The knowledge base in practice

Box 1. A question answered through the knowledge base.

User. https://github.com/AgrDataSci/ai-ready-documentation. We will test four new bean varieties with farmers in Tanzania next season. How many farmers do we need, and can their rankings support a release decision?

Assistant. This answer comes from the tricot knowledge base layer, which holds the published evidence. The handbook by Rachel Chase and colleagues (Chase et al. 2026) sets the rule. Every variety needs at least 30 replicates. Each farmer tests three options. With four varieties that means 120 plots, or 40 farmers. I worked that figure out from the handbook's rule. No paper tests your exact design. The handbook also asks for at least 10 replicates per location and 10 to 20 per geographic area if you want to see location effects. Steinke et al. (2017) assumed up to 20% attrition in their bean work in Honduras. So plan for about 50 farmers per area you want to report on, and more if you will compare women and men. The 150 to 200 farmers you often see quoted are for 10 to 12 varieties. They are not a universal minimum.

Rankings alone will not carry a release decision. De Sousa et al. (2024) report that release authorities want absolute yields, not ranks. Teams therefore combine rankings with yield measured on some plots or by the farmers themselves. Nabateregga et al. (2025) tested this with common bean in northern Tanzania. Farmer measurements with sado tins matched technician measurements closely ($r = 0.96$), and 25 farmers were already enough for accurate estimates. That study did not show that a release committee accepts farmer-measured yield, so do not claim that. Two precedents exist. Daudi et al. (2026) combined station trials in 18 environments with tricot to support groundnut release decisions in Tanzania. IITA released a cowpea variety in Nigeria on tricot evidence. Neither one shows that Tanzanian authorities formally accept tricot data for beans. Confirm that with the release committee before you design the trial.

The user shared the link of ai-ready-documentation and asked the question. The assistant (Claude Opus 5.5, 31 October 2026) read the routing skill, which sent it to the knowledge base, then opened six cards and the people file. The answer is reproduced unedited. Every number in it was checked against the cards.

Box 1 shows the behaviours the knowledge base is designed to produce. The assistant answers the question directly and grounds each number in a named source. It says when a figure is its own calculation rather than a published result, and it keeps each study's crop, country and design. It separates farmer rankings from measured yield, and a precedent from formal acceptance, so the user does not overstate what the evidence supports. It credits the people behind the work by name. The same answer, given from general training data, could look just as fluent while resting on none of this.

Place in the AI-ready suite

The knowledge base is the evidence layer of ai-ready-documentation (`https://github.com/AgrDataSci/ai-ready-documentation`), a set of linked repositories in which each layer carries its own skill. The other layers hold the field protocols, the harmonized on-farm trial data and the team's analytical approaches, and a planned layer will link the trial data with climate, soil and other environmental data. A shared template supports reproducible analyses across them (de Sousa and Laporte, 2026). The suite and its design are described in the ai-ready-documentation repository.

The layers connect through shared identifiers. A card names the dataset its findings rest on, a dataset names the protocol that produced it, and protocols and data share trait definitions from the Crop Ontology. An assistant can follow these links from a published finding to the trials behind it and, once the integration layer is in place, to the climate data that help explain it.

Content

The knowledge base holds one card for each source document, covering fifteen years of published evidence, from the first proposal of the approach (van Etten, 2011) to the current tricot handbook (Chase et al., 2026). Most sources concern tricot, but the knowledge base also covers related work on product testing with users, such as consumer testing of processed foods, product concept testing with videos, and data standards for breeding. By type, the majority are peer-reviewed research articles, and the others include software papers, reviews, guides, reports, policy briefs, working papers, preprints and presentations, each labelled by type so that an assistant can weigh it accordingly. The repository grows as new work is published, and each change updates the version of the skill. The cards cover pre-breeding with genebanks, the origins and guides of the approach, data quality and impact, climate and genomic analysis, software and data standards, applications to specific crops, consumer and product concept testing, gender and inclusion, and the mainstreaming of tricot in breeding programs.

Each card follows one template. Its metadata give the DOI or repository handle, a stable open copy, the license, the dataset and code behind the paper, the crops and regions covered, search keywords and a one-sentence gist. Its body lists the questions the source answers, a summary of 400 to 500 words written in our own words, the methods in brief, the limitations stated by the authors, a table of key numbers with their primary sources, and notes on discrepancies with other cards. An optional field records claims the source is likely to be stretched to but does not support.

Repository structure

```
tricot-knowledge-base/
|-- SKILL.md            # Skill instructions, version and paper index
|-- people.md           # People behind the papers, naming and consent
|-- AGENTS.md           # Entry point for AI agents
|-- llms.txt            # Short map of the repository for AI tools
|-- index.json          # Version, card hashes and links (generated)
|-- relations.jsonld    # Links between cards as JSON-LD (generated)
|-- papers/             # One card per source
|-- papers-pdf/         # Source documents and their licenses
|-- TEMPLATE.md         # Template for new cards
|-- scripts/            # Rebuilds the index and version
|-- tests/              # Evaluations of card format and answers
|-- docs/               # This paper
|-- CITATION.cff
|-- LICENSE
```

Key features

- **Verified numbers.** Every number in a card is checked against the full text of its source. Disagreements between a paper's text and tables, or between papers, are recorded in the card notes rather than smoothed over.
- **Primary sources.** When a paper reports a figure from another study, the card names that study, so users cite the original.
- **Linked papers.** Cards declare how they build on each other with terms from the Citation Typing Ontology (Peroni and Shotton, 2012), such as `extends`, `usesDataFrom` and `disagreesWith`. A script checks every link, adds the inverse links and exports the graph as JSON-LD, so an assistant can follow a line of work from one paper to the next.
- **Evidence boundaries.** The skill asks the assistant to separate direct evidence, indirect evidence, methodological inference and evidence gaps. It must keep each study's crop, country, sample and design when it transfers a finding, and never present a study-specific sample size as a universal rule.
- **People and consent.** A people file records how to credit the researchers behind each card and whether they agreed to be suggested as contacts. The assistant credits work by name but never speaks for a person, suggests a contact only with recorded consent and through an institutional channel, and never shares personal contact details or photos.
- **Versioning and updates.** A script computes a version from the content of every card and writes it to the index. A GitHub Action rebuilds the index and publishes the packaged skill on every change. Once per conversation, an installed skill compares its version with the live repository and fetches only what changed.
- **Light skill.** The assistant loads a short index first and opens only the cards a question needs, usually two to five. The packaged skill stays small, so it travels easily between users.
- **Open and readable.** Cards are plain markdown under a CC BY 4.0 license. Source documents keep their original licenses, which are listed for every file.

Verification and evaluation

We tested two card formats on the same source. A short card held a summary and key numbers. A long card added the questions the paper answers, the methods and the stated limitations. Answering expert questions from one card at a time, the long format gave more complete and better-grounded answers, for example the reasoning behind a recommended trial size rather than the number alone. All cards now use the long format.

A second evaluation probes whether an assistant keeps evidence boundaries, regulatory distinctions, primary-source provenance and contact consent. It poses adversarial questions, such as whether tricot is approved for releasing a crop in a country without a primary record, or whether a researcher's email can be found online when consent is pending. Answers are scored on factual accuracy, provenance, scope, uncertainty and consent. Any fabricated source, unsupported claim of regulatory approval or breach of consent counts as an automatic failure. The evaluation is meant to be rerun after every change to the skill.

Supporting demand-led product development

The knowledge base answers evidence questions across the life of a product. Before a product exists, users can ask how concept testing has helped define product profiles and market segments. While designing a trial, they can ask how many participants and entries are needed, how to combine rankings with yield measured on a subset of plots, or how to sample so that women and different user segments are represented. When reading results, they can ask how to interpret rankings across environments, how climate shapes performance, or which traits drive consumer acceptance. When deciding which candidates to advance or release, they can ask what evidence release committees have accepted. Each answer comes from verified cards, cites the primary source, keeps the context of the original study, and says where the evidence stops.

Through the data layer, the same skills let an assistant move from published evidence to the trial data of a specific crop and region, and the planned integration layer will add agroclimatic information. Together, these resources form the foundation for AI-supported recommendations that match candidate products with user and market demand and with climate risk.

The rules in the skill also serve as practical guidance for responsible use. They separate measured yield from rankings, candidate advancement from official release, and direct evidence from extrapolation. They also protect the people behind the work. The same cards can support training in demand-led product testing for researchers, technicians and students.

Implementation and usage

Users reach the knowledge base in four ways. The simplest is to share one link.

1. **In Claude.** Download the latest skill package from the repository releases and upload it as a skill. The assistant then consults the index whenever a question touches tricot or on-farm testing.
2. **In Claude Code.** Clone the repository into the skills directory.
3. **By sharing a link.** Paste the link of ai-ready-documentation (`https://github.com/AgrDataSci/ai-ready-documentation`) into an AI assistant with web access, as in Box 1. Its routing skill sends each question to the right layer, here the knowledge base, and the assistant answers from the live files. The link of the knowledge base itself works the same way for evidence questions.
4. **By hand or with other tools.** Start from the index table in `SKILL.md` and open the relevant cards in `papers/`.

Contributors add a source by extracting its text, drafting a card from the template, checking every number against the source, adding links to related cards and authors, and opening a pull request. The index and version update automatically when the change is merged.

Limitations

The knowledge base covers openly available sources produced by or with the tricot team, so it reflects the perspective of the people who developed the approach. It is a curated synthesis, not a systematic review, and the absence of a topic means it is absent from these sources, not from the literature. Summaries are written by the team and checked against the sources, but they remain interpretations, and users should cite and consult the original papers. The knowledge base informs decisions about products but does not make them. Recommendations for specific candidates require the trial data, environmental data and analyses of the other layers, several of which are still under development. Answers depend on the assistant following the rules in the skill, which the evaluations test but cannot guarantee. Consent to be suggested as a contact is collected person by person, so the assistant suggests contacts only for those who have agreed.

Conclusion

Demand-led products need evidence on what users value, how products perform in their hands, and how climate shapes that performance, at every stage from concept to release. Curated, verified and openly licensed evidence lets AI assistants deliver it accurately, credit the people behind the work and say where the evidence stops. The knowledge base grows with the literature, stays current in every installed copy, and connects published findings to the data and analyses that support the development of products people want.

Code and data availability

The knowledge base, its source documents and this paper are openly available on GitHub at `https://github.com/AgrDataSci/tricot-knowledge-base`. The cards and documentation are released under a CC BY 4.0 license. Source documents keep their original licenses, listed in `papers-pdf/LICENSES.md`.

Acknowledgments

This work is part of the CGIAR Digital Transformation Accelerator. We thank all funders who supported this research through their contributions to the CGIAR Trust Fund (`https://www.cgiar.org/funders/`). [Additional grants to be confirmed.] We thank the authors of the papers in the knowledge base.

References

R. Chase, K. de Sousa, M.-A. Laporte, et al. *Product testing with users in agriculture and food. Decentralized on-farm evaluation, breeding product validation, consumer testing and other applications using the tricot approach*. Bioversity International, Montpellier, France, 1st edition, 2026. doi: 10.5281/zenodo.20042955.

K. de Sousa and M.-A. Laporte. A FAIR and project-oriented template for open science data workflows. Zenodo, 2026. URL `https://doi.org/10.5281/zenodo.15224388`. Version 1.3.

K. de Sousa, J. van Etten, J. Poland, et al. Data-driven decentralized breeding increases prediction accuracy in a challenging crop production environment. *Communications Biology*, 4:944, 2021. doi: 10.1038/s42003-021-02463-w.

K. de Sousa, J. van Etten, R. Manners, et al. The tricot approach. An agile framework for decentralized on-farm testing supported by citizen science. A retrospective. *Agronomy for Sustainable Development*, 44:8, 2024. doi: 10.1007/s13593-023-00937-1.

Silvio Peroni and David Shotton. FaBiO and CiTO. Ontologies for describing bibliographic resources and citations. *Journal of Web Semantics*, 17:33–43, 2012. doi: 10.1016/j.websem.2012.08.001.

J. van Etten, E. Beza, L. Calderer, K. van Duijvendijk, C. Fadda, B. Fantahun, Y. G. Kidane, J. van de Gevel, A. Gupta, D. K. Mengistu, D. Kiambi, P. N. Mathur, L. Mercado, S. Mittra, M. J. Mollel, J. C. Rosas, J. Steinke, J. G. Suchini, and K. S. Zimmerer. First experiences with a novel farmer citizen science approach. Crowdsourcing participatory variety selection through on-farm triadic comparisons of technologies (tricot). *Experimental Agriculture*, 55(S1):275–296, 2019. doi: 10.1017/S0014479716000739.

Jacob van Etten. Crowdsourcing crop improvement in sub-Saharan Africa. A proposal for a scalable and inclusive approach to food security. *IDS Bulletin*, 42(4):102–110, 2011. doi: 10.1111/j.1759-5436.2011.00240.x.

R. C. Voss, K. de Sousa, S. N'Danikou, et al. Citizen science informs demand-driven breeding of opportunity crops. *Plants, People, Planet*, page e70035, 2025. doi: 10.1002/ppp3.70035.